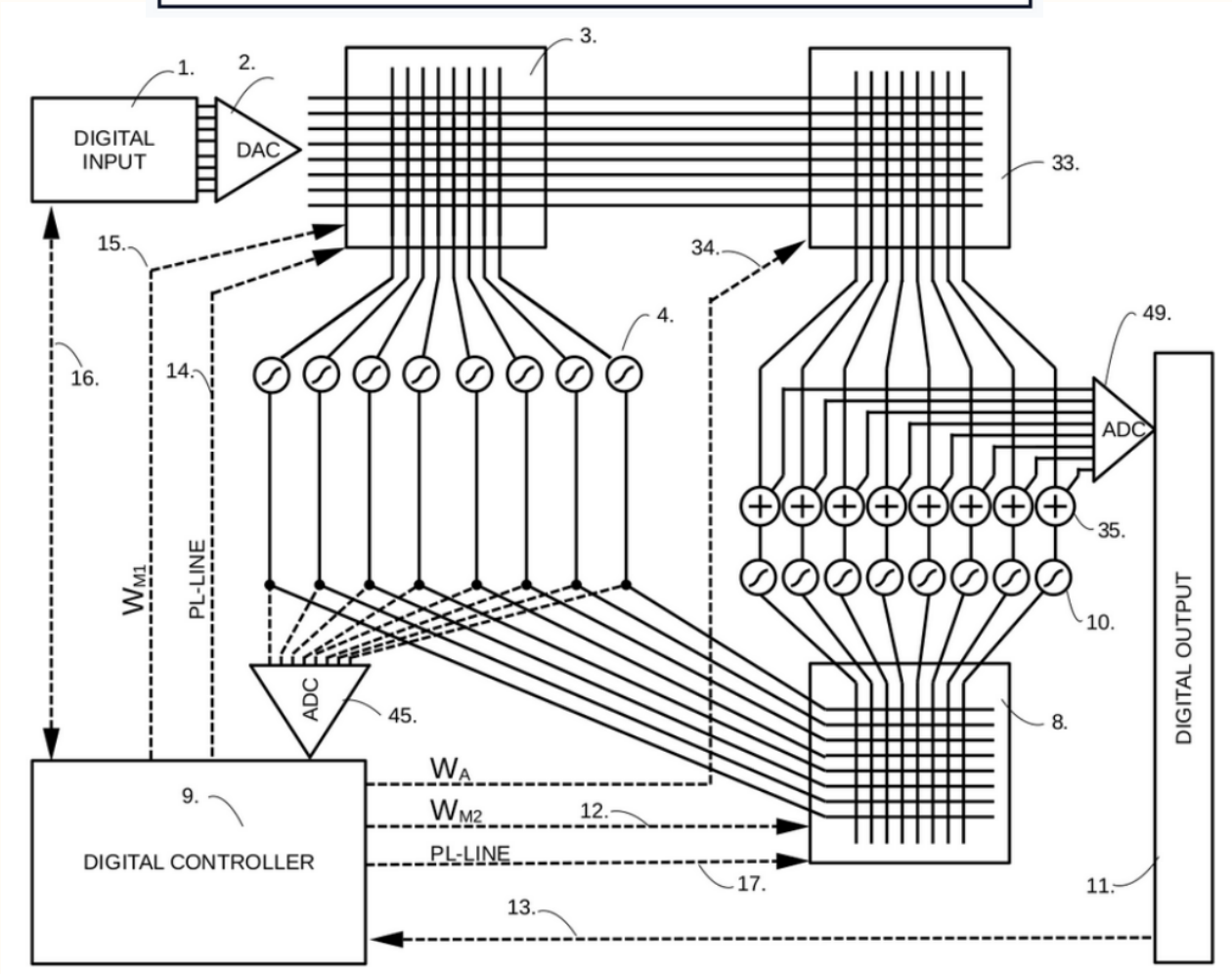
TITLE: INVERTED MAML FOR ANALOG IMC MESH

PATENT APP NO: SE 2630397-4

STATUS: PATENT PENDING

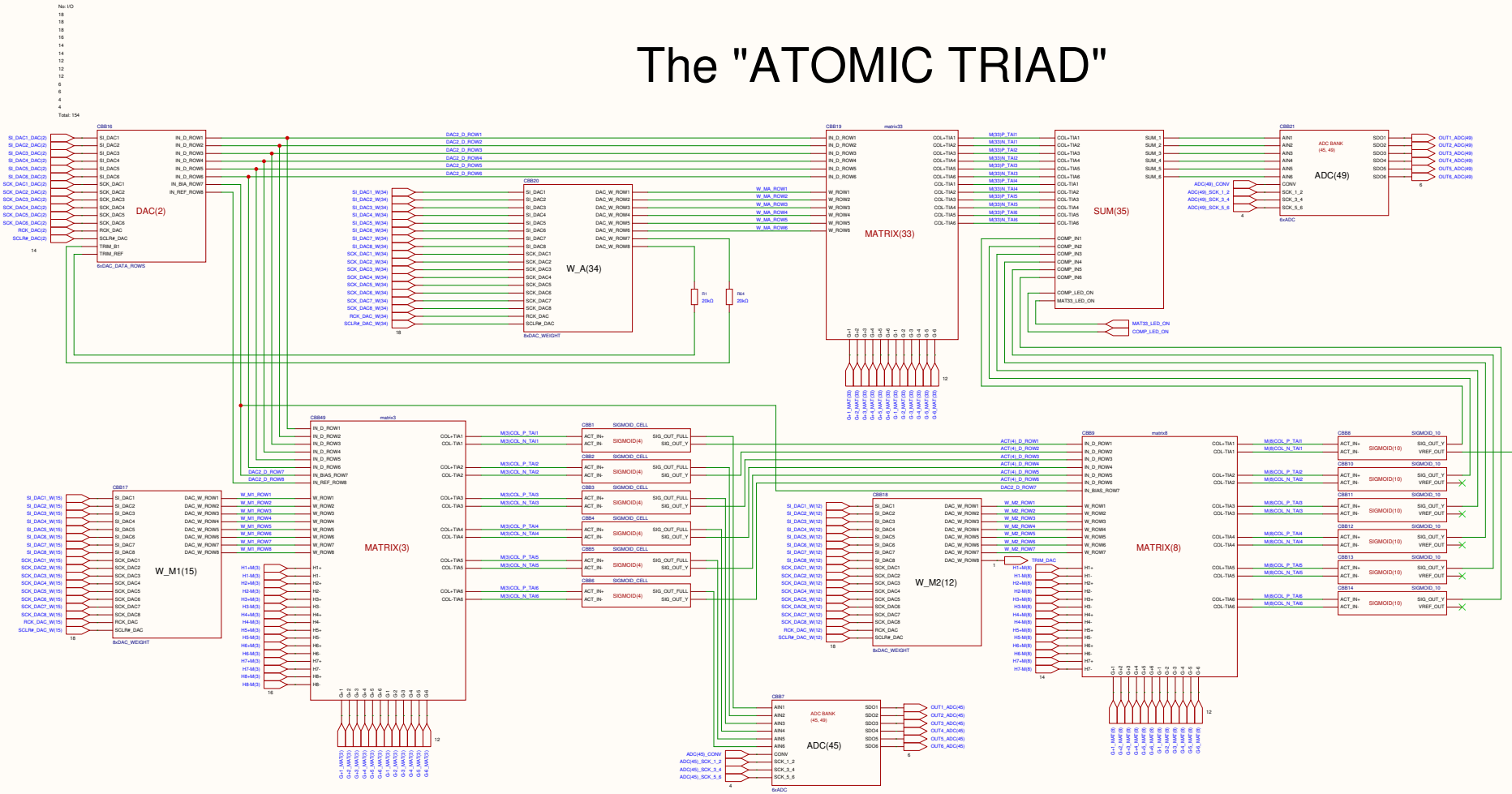
INVENTOR: OLLE WELIN

PRIORITY DATE: 2026-06-03



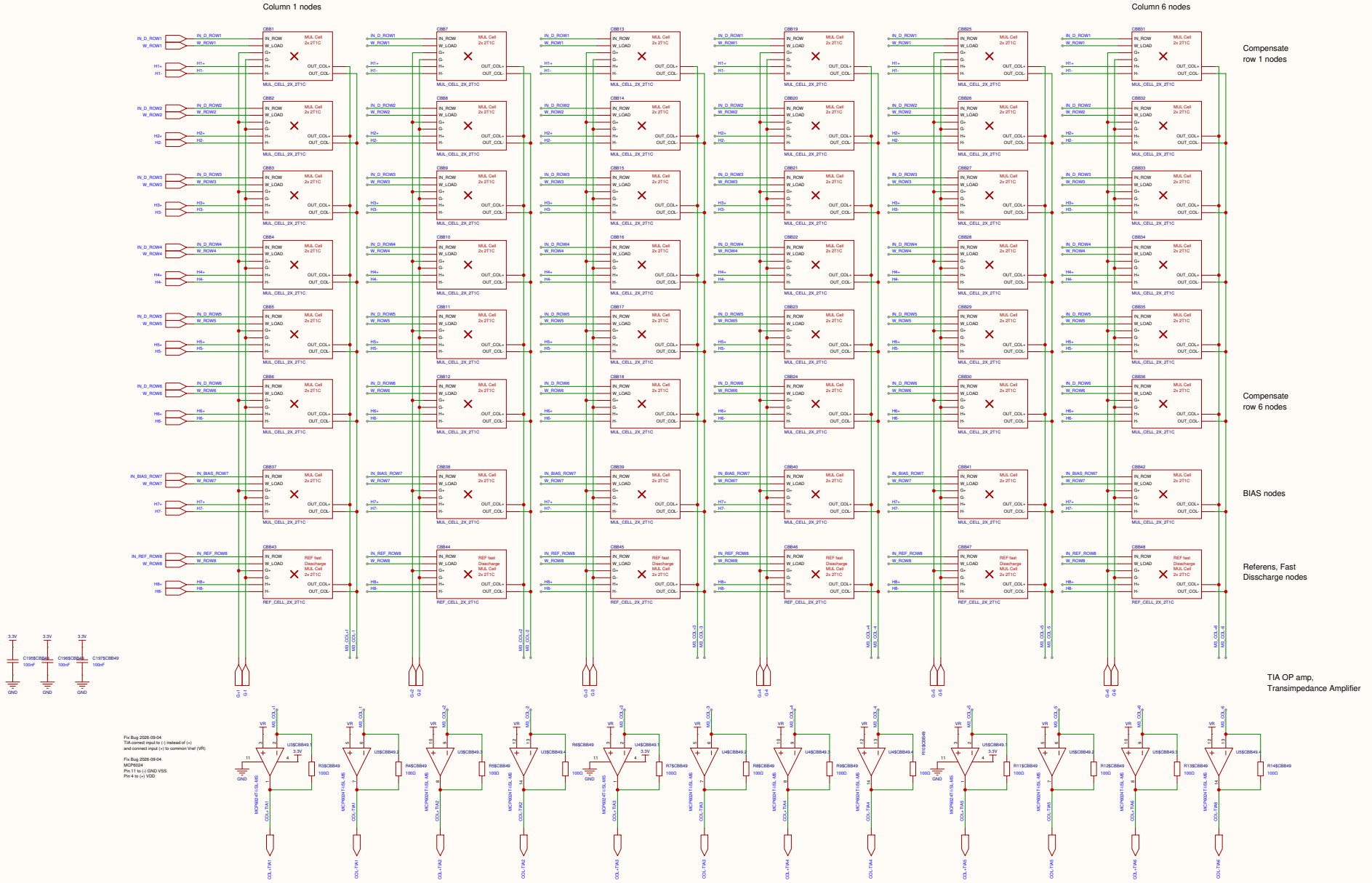
Schematic	Inverted_MAML	Create at	2026-06-19
		Update at	2026-06-19
Board	Analog_Matrix	Page	cover
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
	Version	Size	Page 1 Total 5
EasyEDA	V1.0	A4	EasyEDA.com

The "ATOMIC TRIAD"



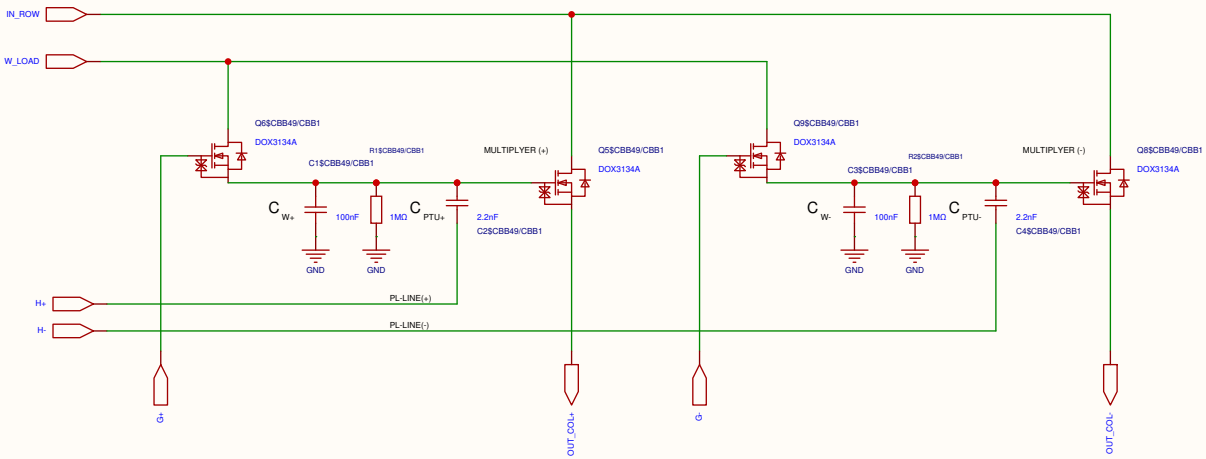
Schematic	Inverted_MAML	Create at	2026-07-19
		Update at	2026-09-04
Board	Analog_Matrix	Page	Atomic_Triad
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 2 Total 5
EasyEDA		V1.0	A4	EasyEDA.com

$$V_{ut_TIA} = 1.65\ \text{V} - (5\ \text{mA} \times 100\ \Omega) = 1.65\ \text{V} - 0.5\ \text{V} = \mathbf{1.15\ V}$$



Schematic	matrix3	Create at	2026-06-27
Board		Update at	2026-09-04
Drawn		Page	matx3
Reviewed			
		Inverted_MAML_6x6	
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

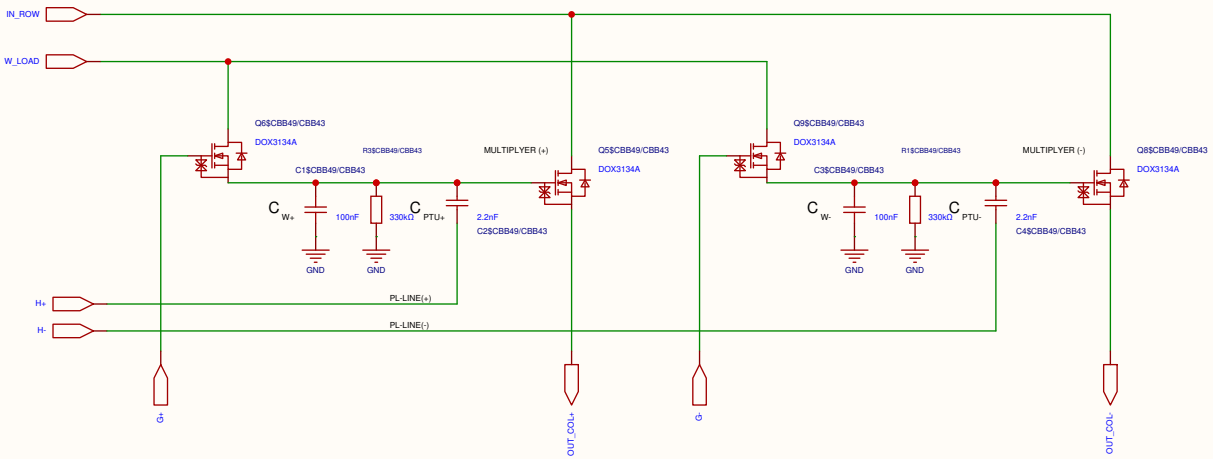
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-07-18
		Update at	2026-07-18
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size / Page 1 Total 1
EasyEDA		V1.0	A4 / EasyEDA.com

Analog REF DISSCHARGE Multiplication CELL Kernel Differential 2x 2T1C



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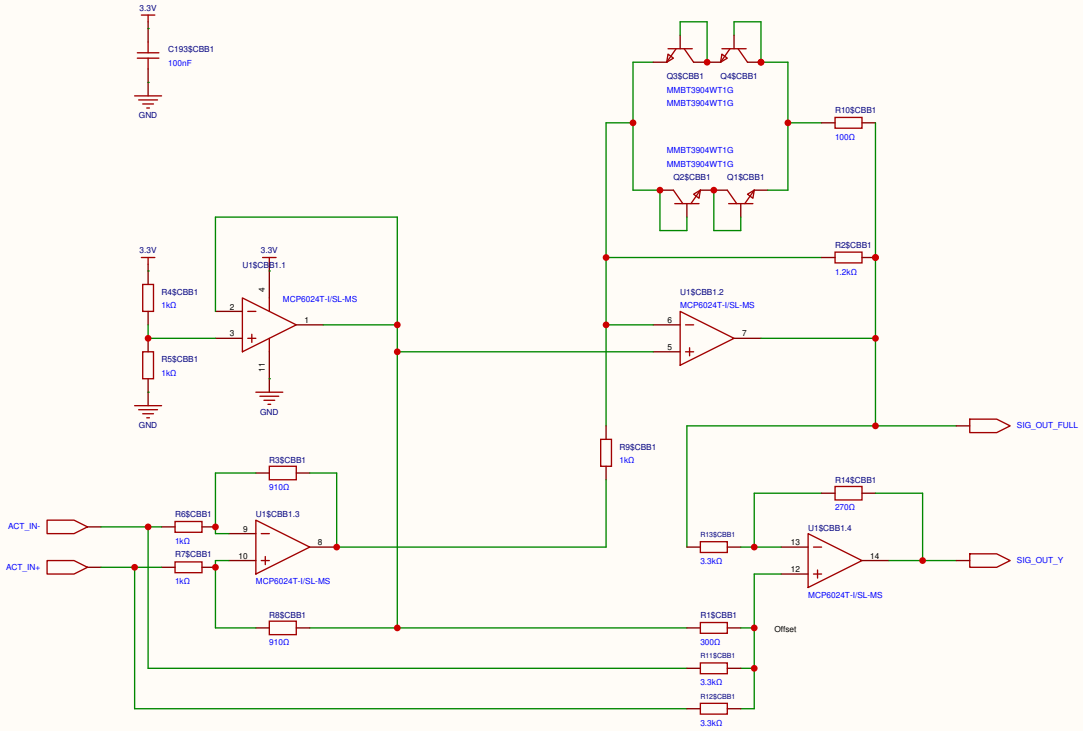
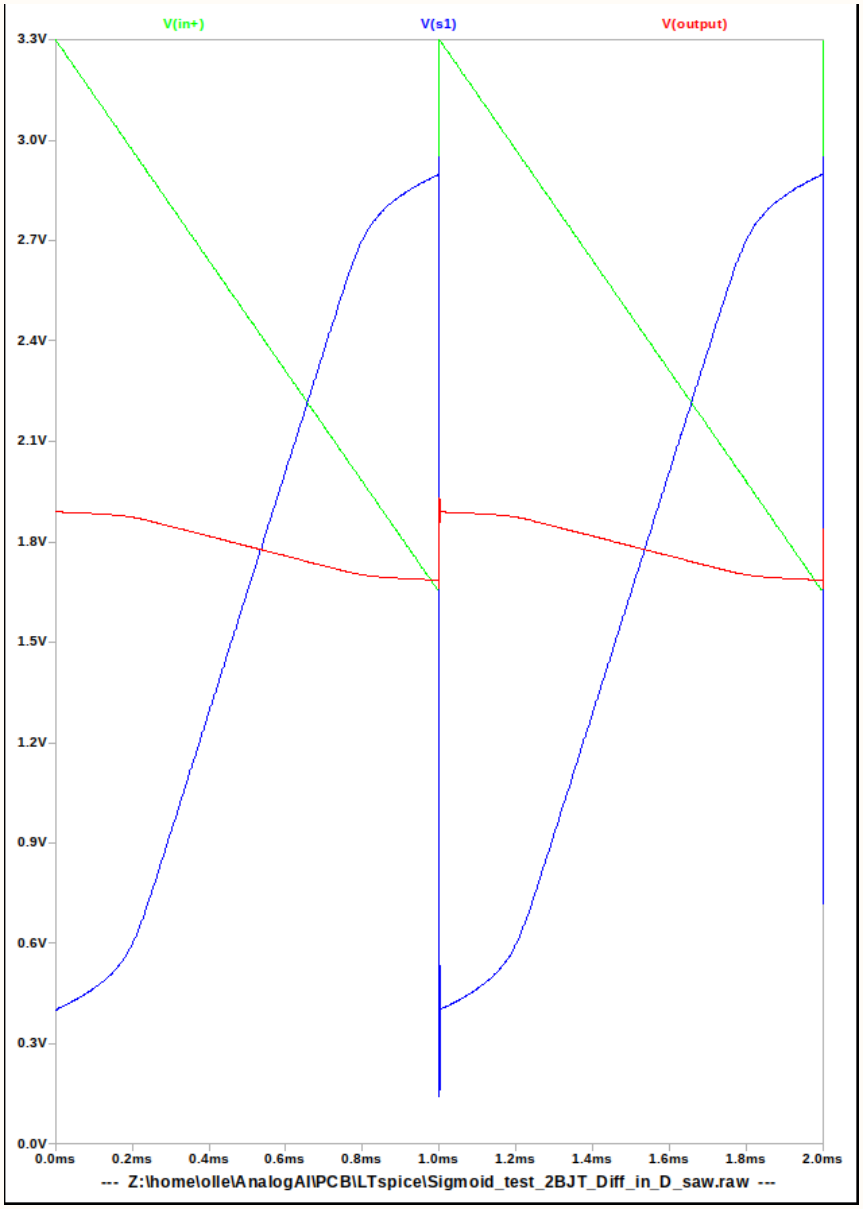
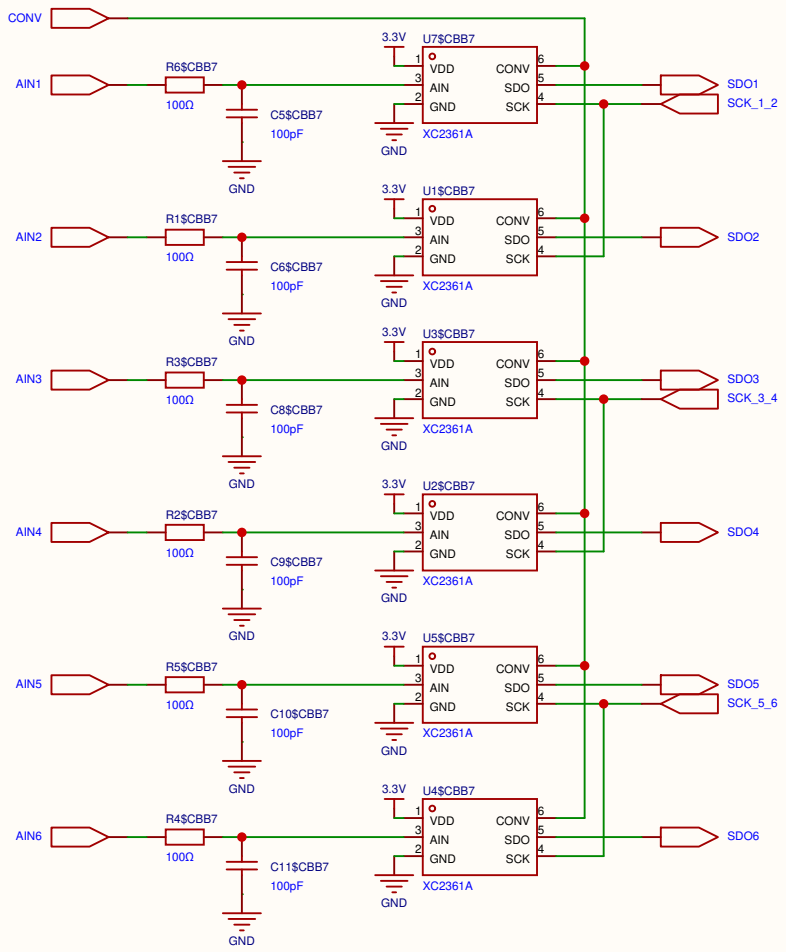
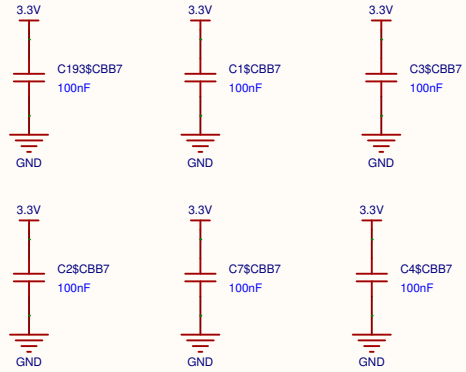
Schematic	REF_CELL_2X_2T1C	Create at	2026-07-18
		Update at	2026-07-18
Board		Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com





Schematic	SIGMOID_CELL	Create at	2026-07-07
		Update at	2026-07-07
Board		Page	SIMOID(4)
Drawn		Inverted_MAML_6x6	
Reviewed			
		Version / Size	Page 1 Total 1
EasyEDA		V1.0 / A4	EasyEDA.com





Schematic	6xADC	Create at	2026-08-11
		Update at	2026-08-11
Board		Page	ADC_BANK
Drawn		Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4
		Page 1 Total 1	
		EasyEDA.com	

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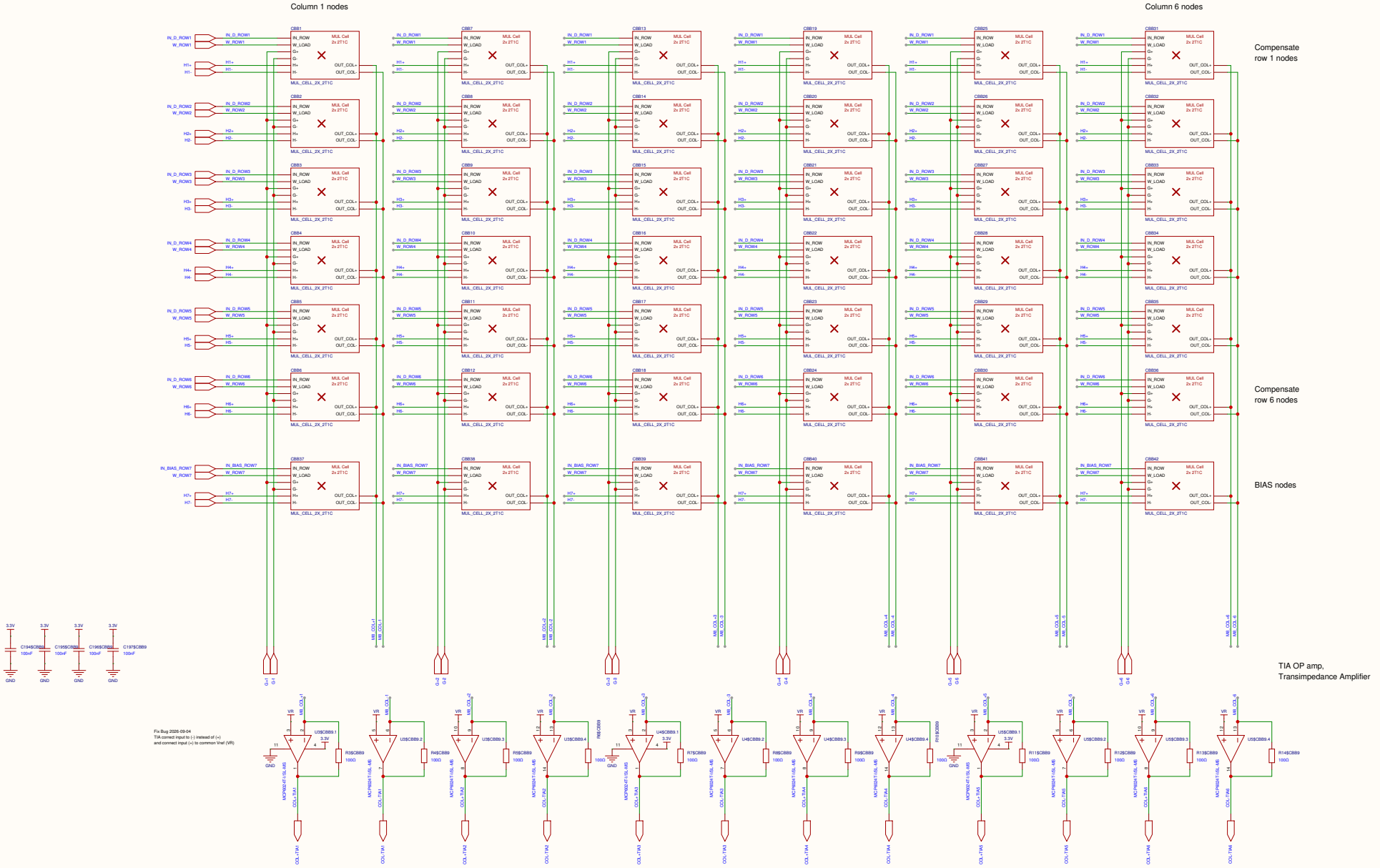
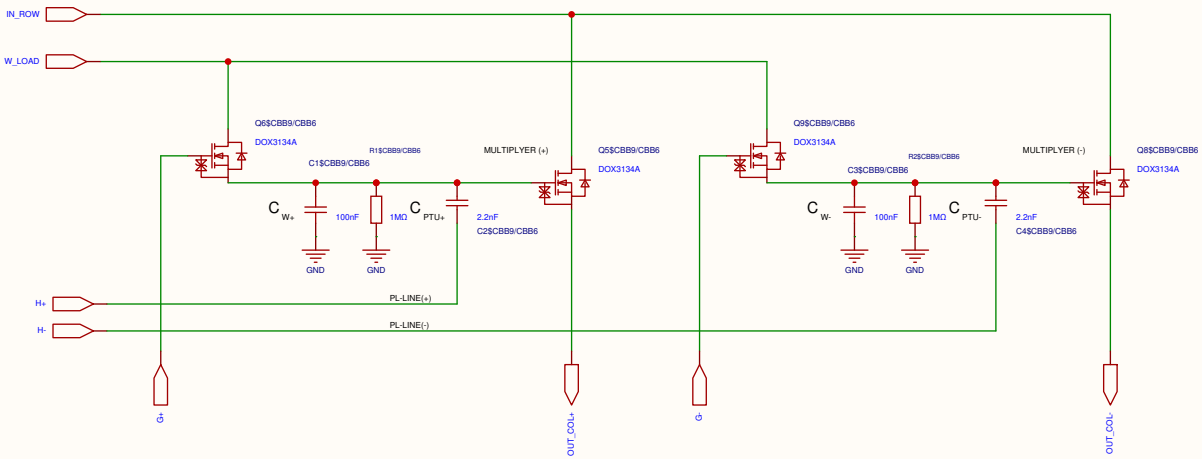


Schematic	matrix8	Create at	2026-06-27
Board		Update at	2026-09-04
Drawn		Page	matx8
Reviewed		Inverted_MAML_6x6	
		Version	Size
		V1.0	A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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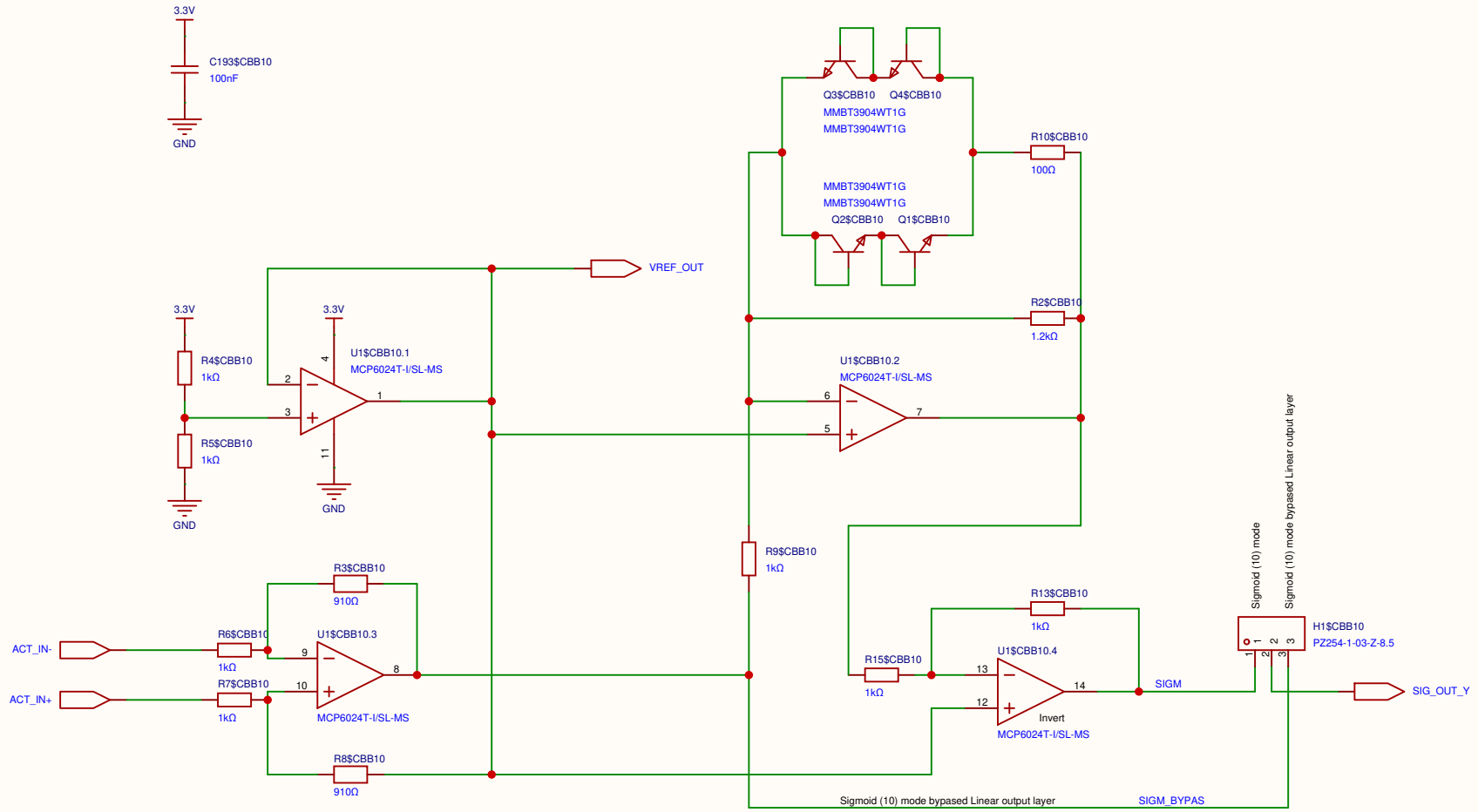
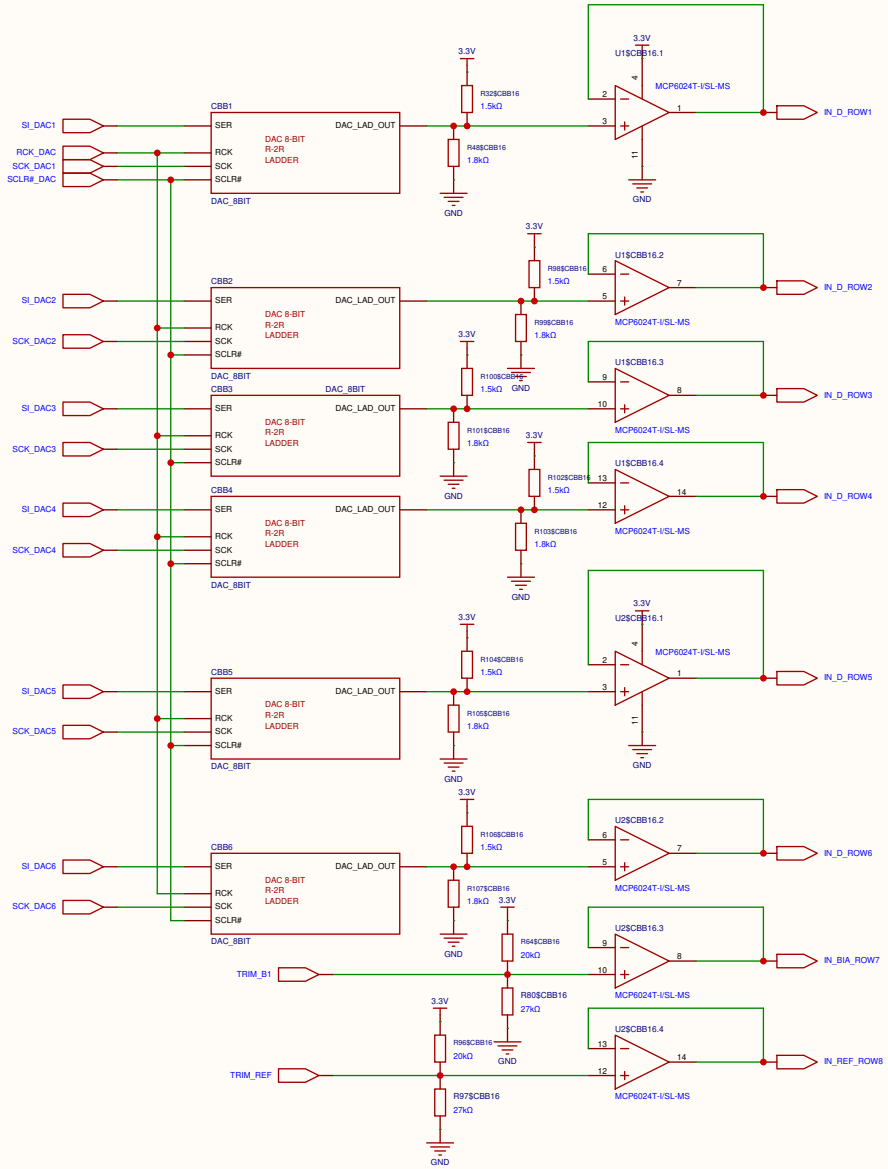
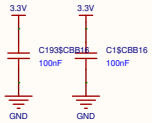
Schematic	MUL_CELL_2X_2T1C	Create at	2026-07-18
		Update at	2026-07-18
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size / Page 1 Total 1
EasyEDA		V1.0	A4 / EasyEDA.com



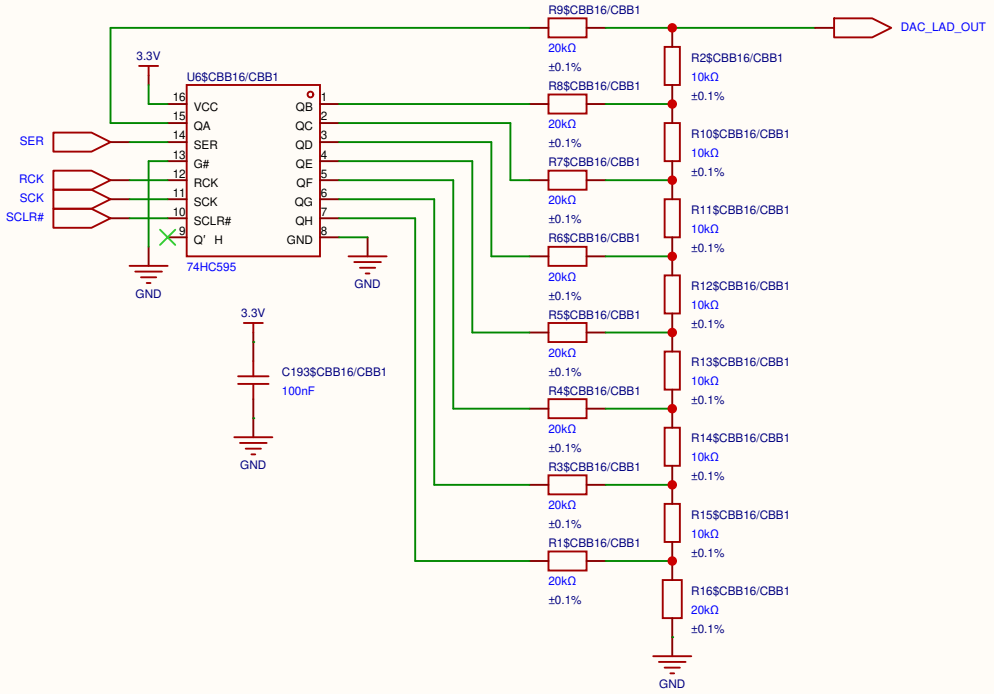
Schematic	SIGMOID_10	Create at	2026-07-18
		Update at	2026-09-04
Board		Page	SIGMOID(10)
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA	V1.0	A4	EasyEDA.com

Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.



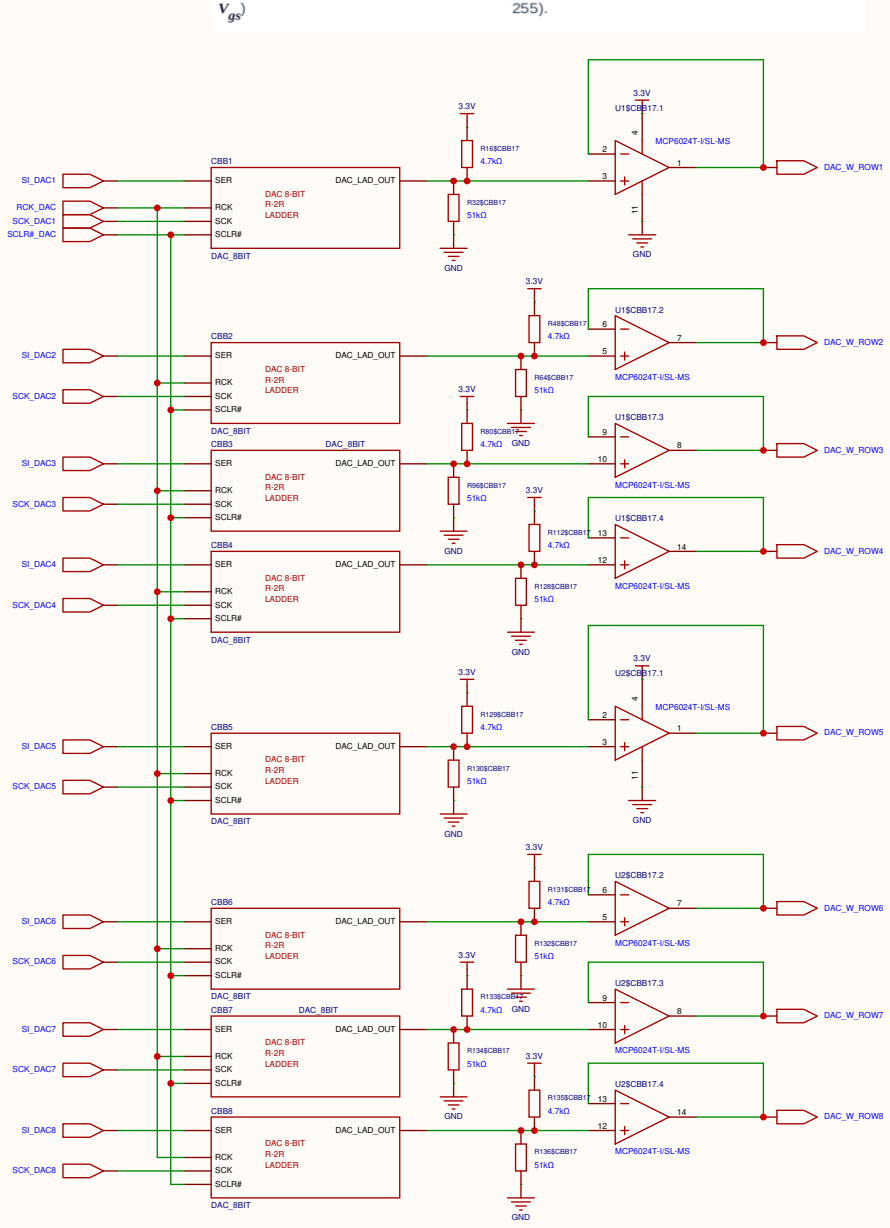


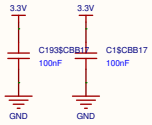
Schematic	6xDAC_DATA_ROWS	Create at	2026-06-20
		Update at	2026-06-20
Board		Page	6xDAC_8-bit
Drawn		Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com



Schematic	DAC_8BIT	Create at	2026-06-19
		Update at	2026-06-19
Board		Page	DAC_8B_R_2R
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
		V1.0	A4
EasyEDA		Page 1 Total 1	
		EasyEDA.com	

Weight swing (Gate / V_{gs}) **1.0 V** Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).





Schematic	8xDAC_WEIGHT	Create at	2026-06-30
		Update at	2026-06-30
Board		Page	8xDAC_8-bit
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

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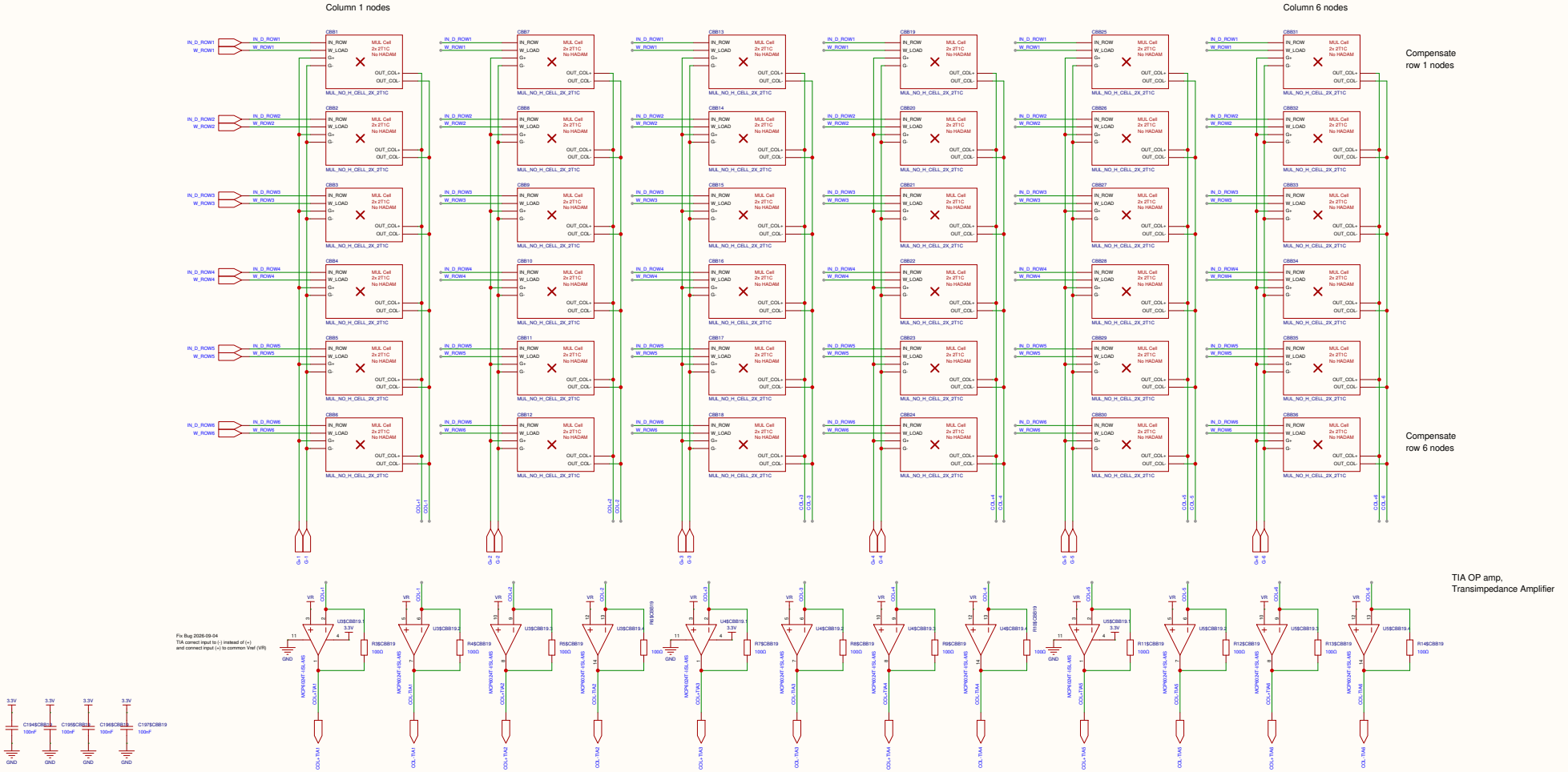
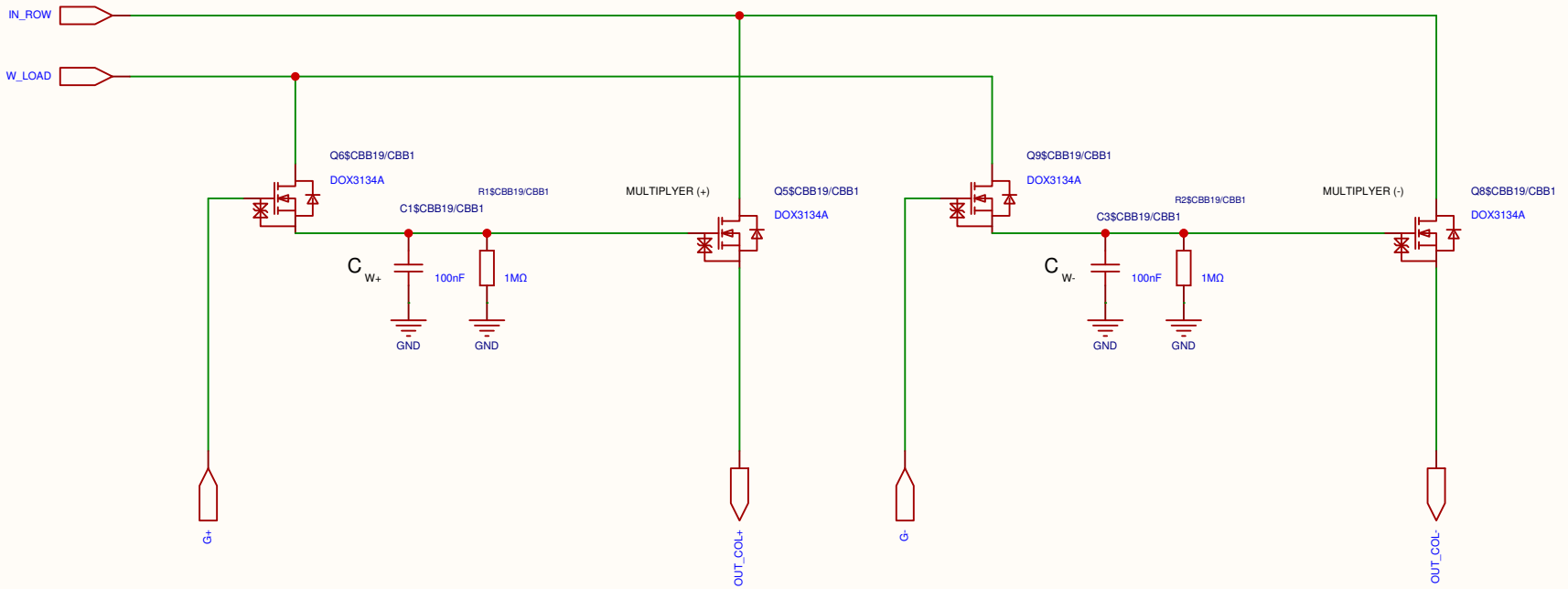


Schematic	matrix33	Create at	2026-06-28
		Update at	2026-09-04
Board		Page	matx33
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

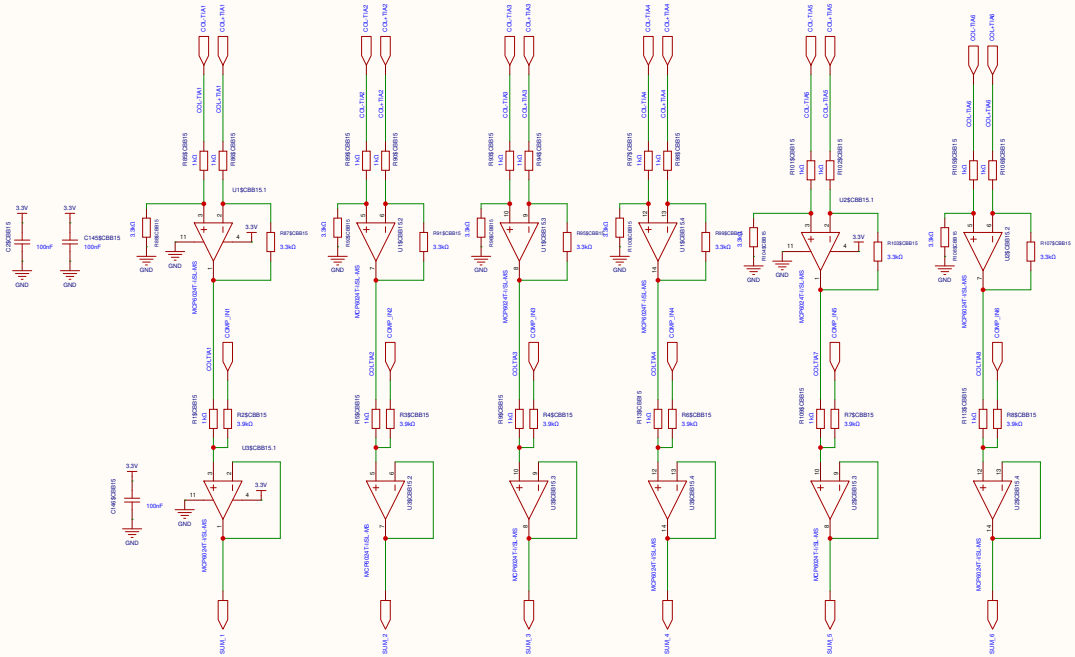
Analog Multiplication CELL Kernel
Differential 2x 2T1C

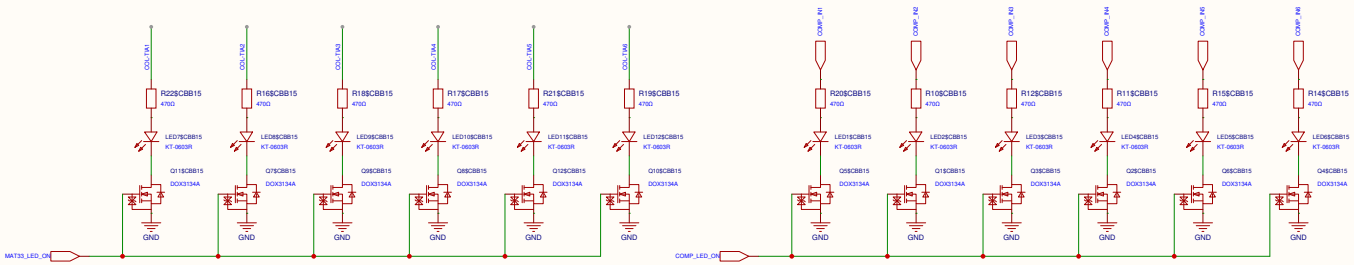


Schematic	MUL_NO_H_CELL_2X_2T1C	Create at	2026-06-28
		Update at	2026-06-28
Board		Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

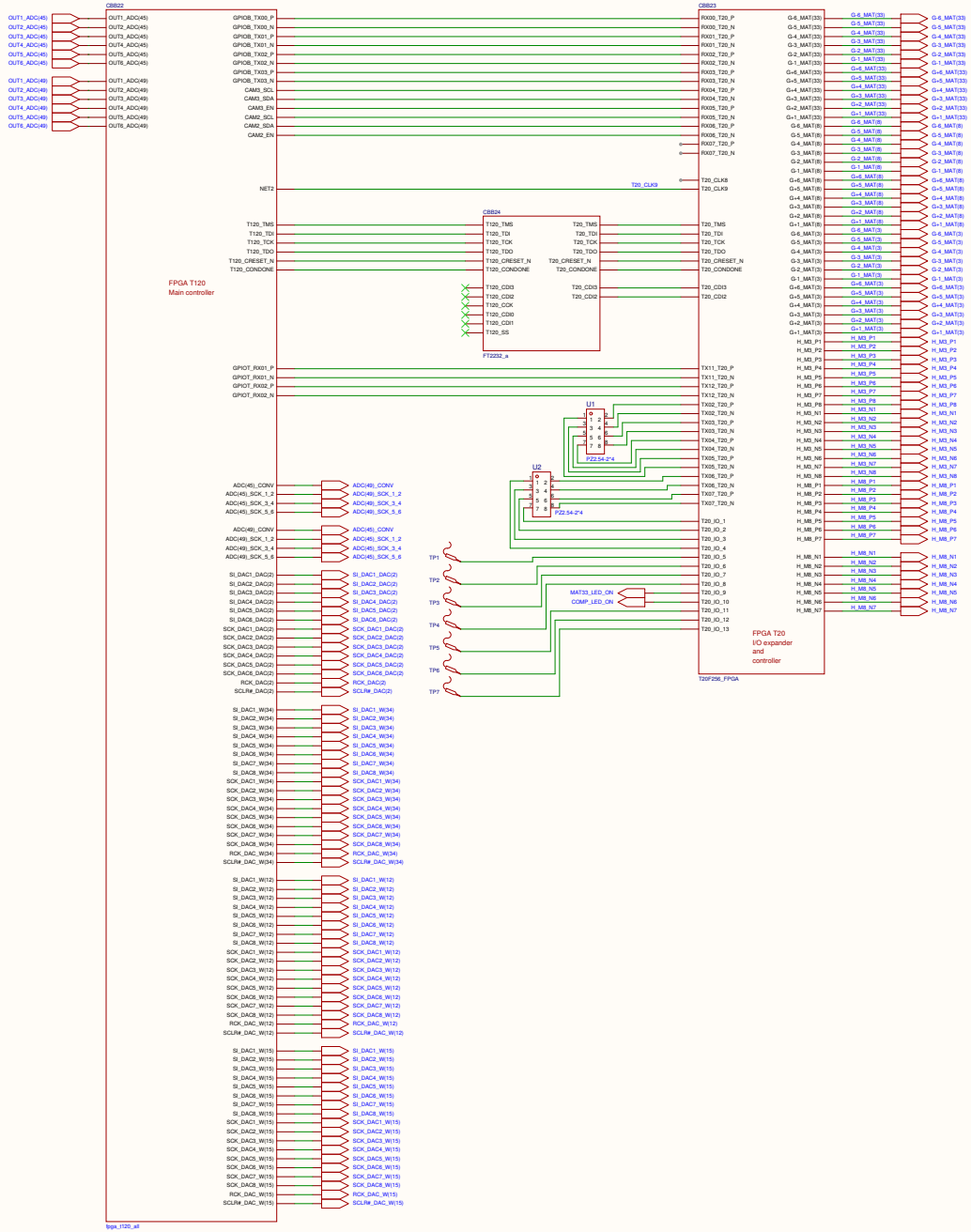
- **Calibration Example:** To match a 3.3 V ADC, the resistors are configured for a Gain of 6.6×.

  $V_{ADC} = 6.6 \times (1.65\ V - V_{out_TIA})$

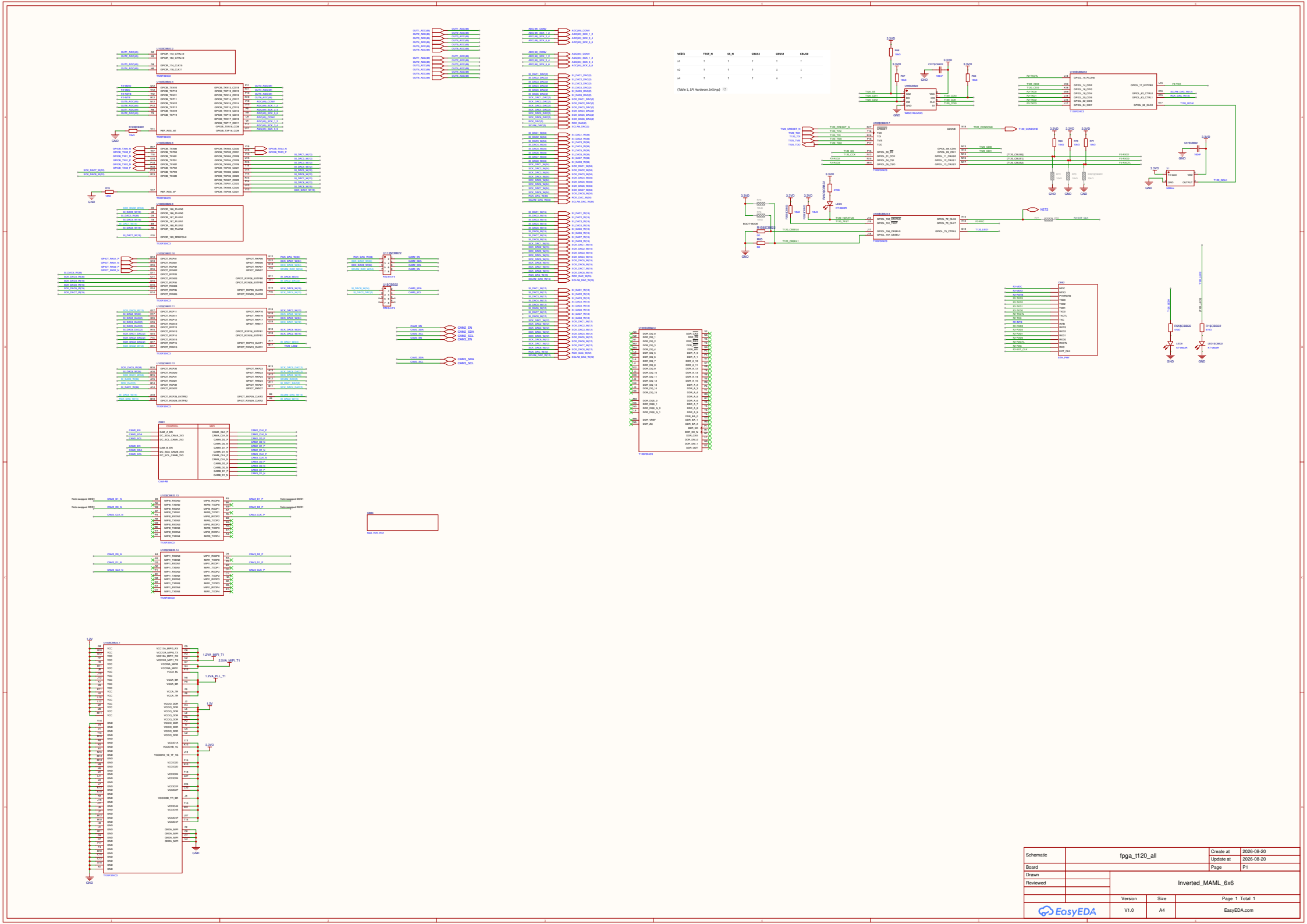




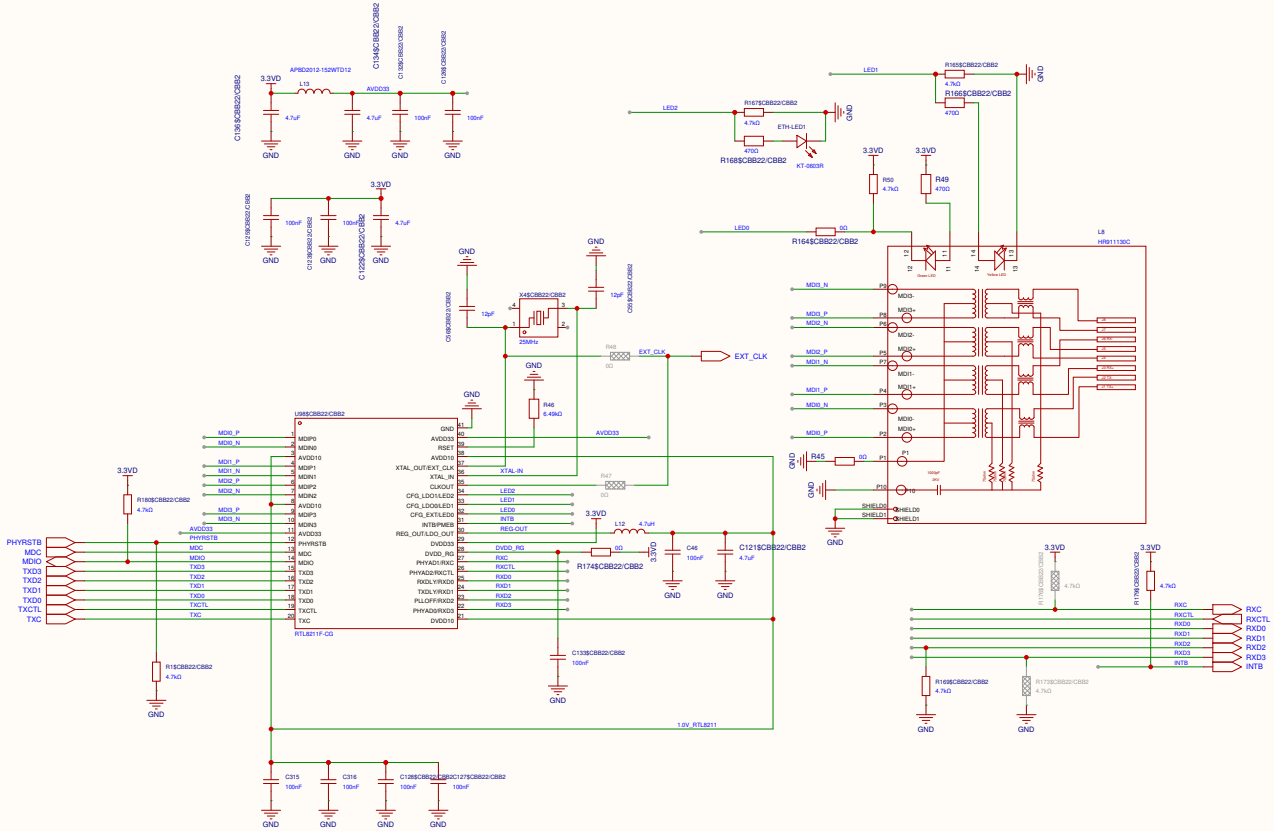
Schematic	SUM35	Create at	2026-08-06
		Update at	2026-09-04
Board		Page	sum35
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com



Schematic	Inverted_MAML	Create at	2026-08-29
		Update at	2026-08-29
Board	Analog_Matrix	Page	Digital_Controller
Drawn		Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 3 Total 5
EasyEDA		V1.0	A4	EasyEDA.com

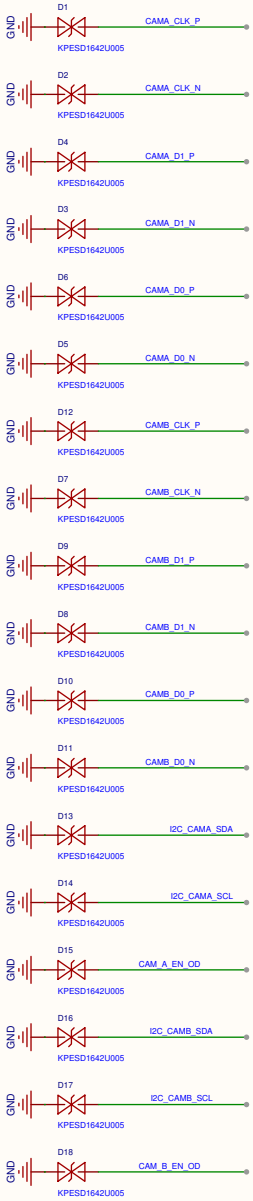


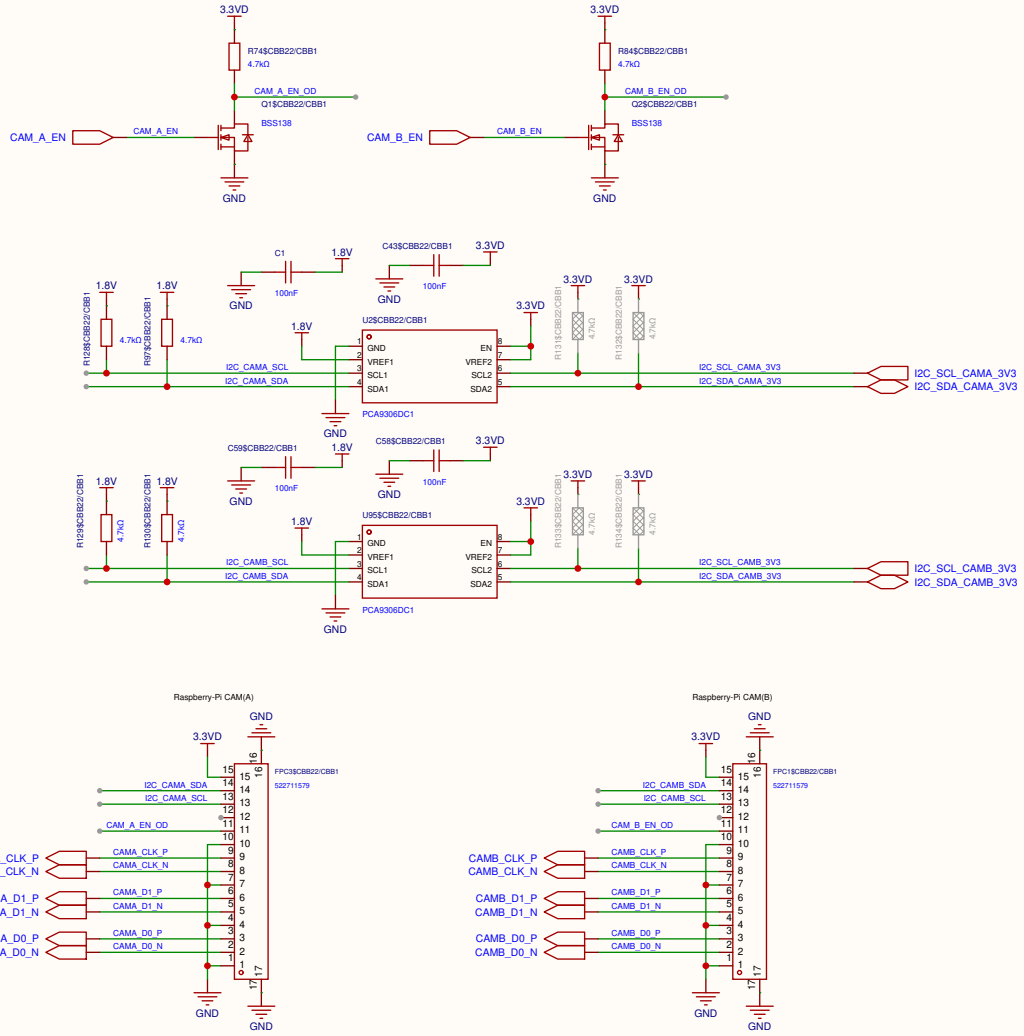
Schematic	fpga_t120_all		Create at	2026-08-20	
Board			Update at	2026-08-20	
Drawn			Page	P1	
Reviewed		Inverted_MAML_6x6			
			Version	Size	Page 1 Total 1
EasyEDA			V1.0	A4	EasyEDA.com



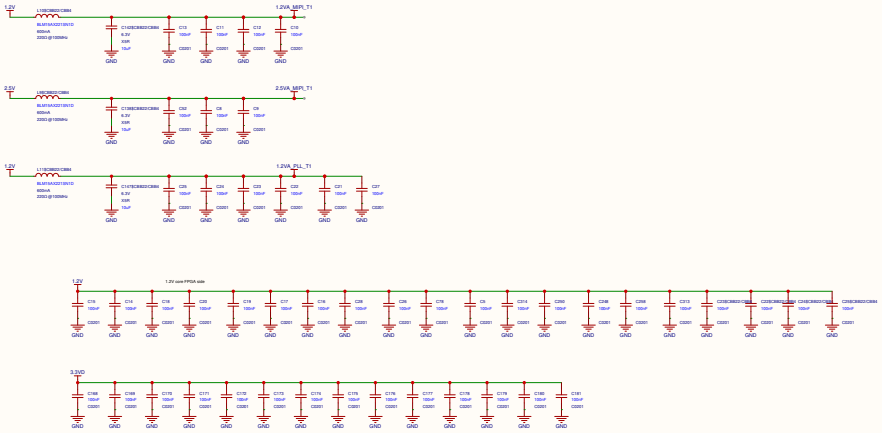
Schematic	ETH_PHY	Create at	2026-08-11
		Update at	2026-08-11
Board		Page	P1
Drawn		Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

CSI- CAM is just for another project I put up this Raspberry camera Interface. (Not for Intverted MAML proof of concept)

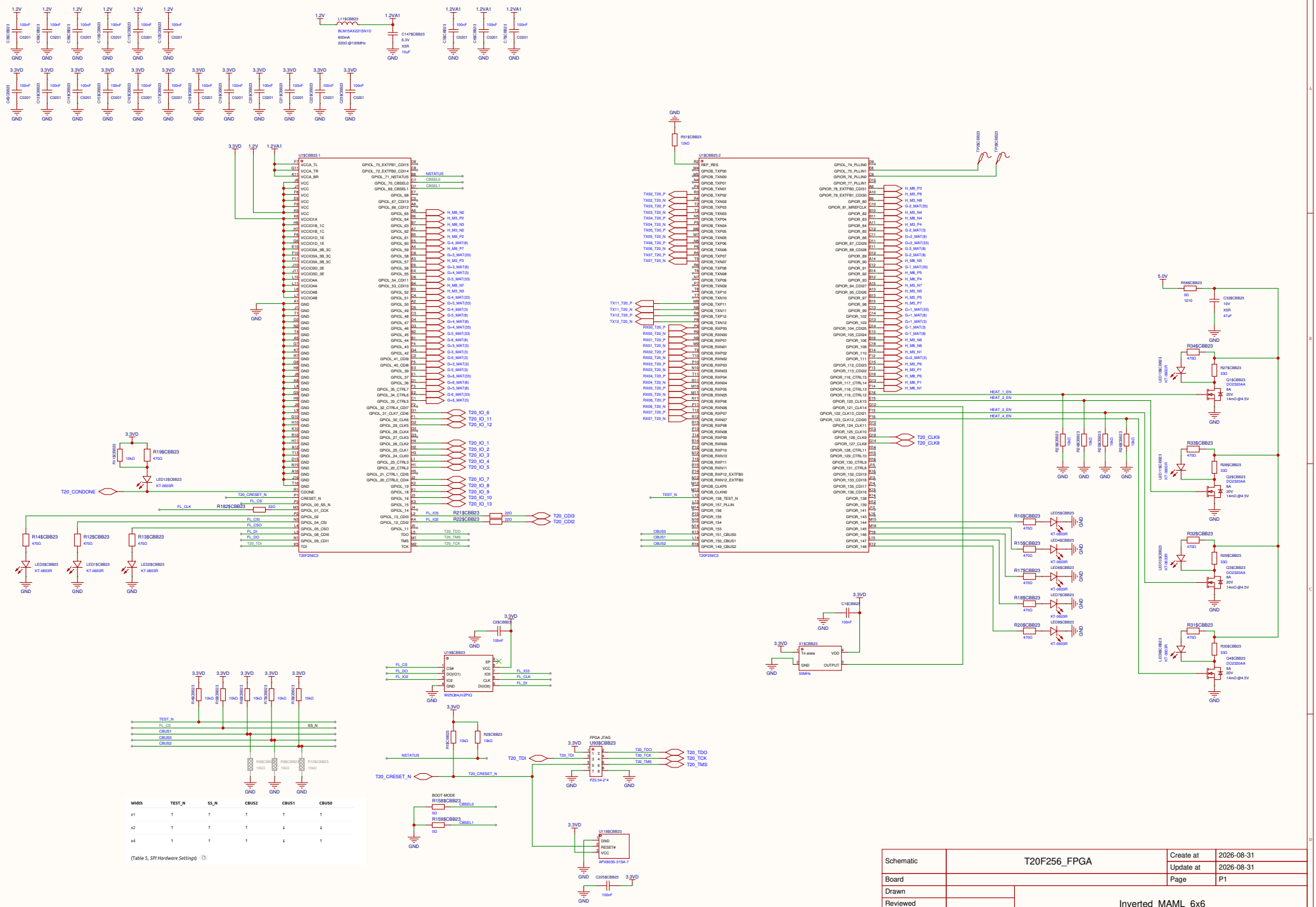


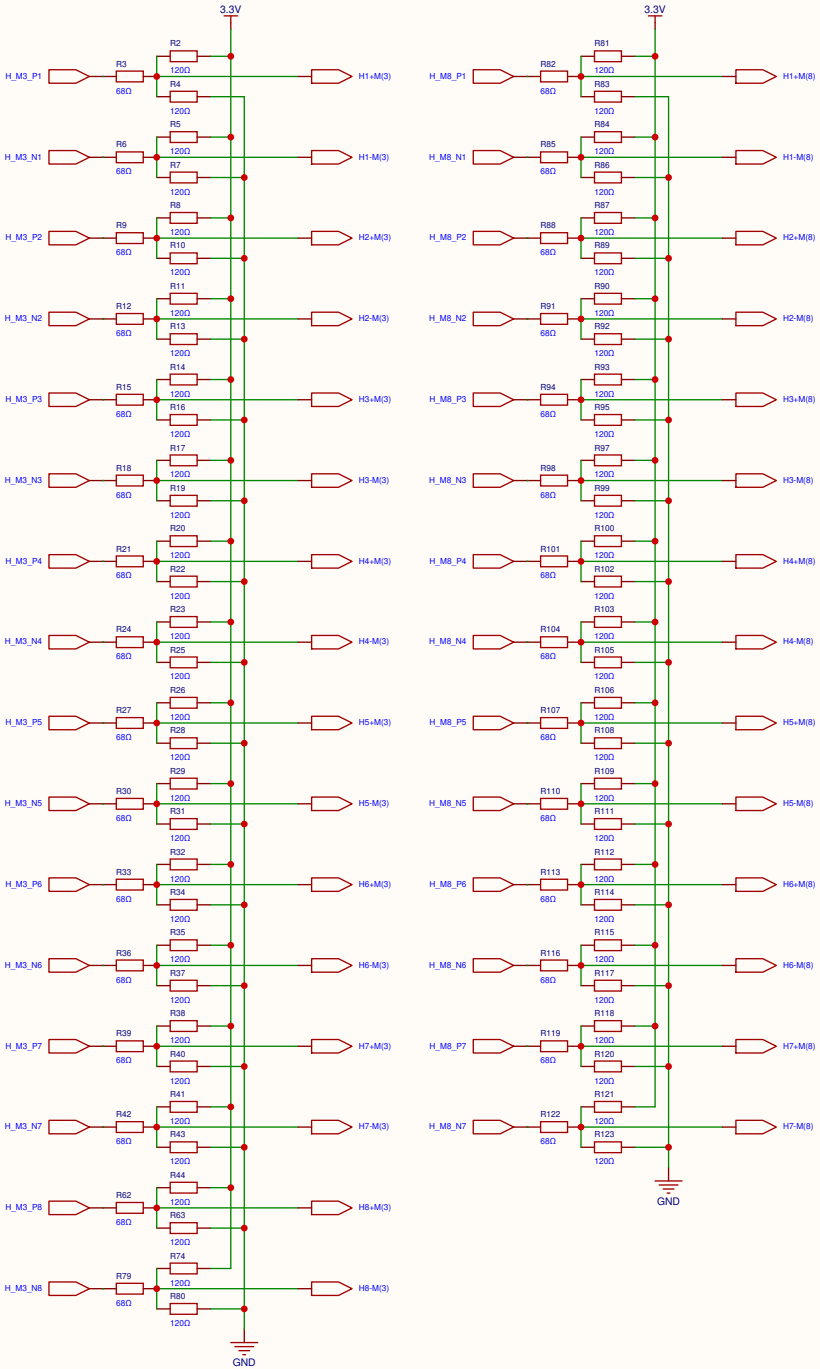


Schematic	CAM-AB	Create at	2026-08-19
		Update at	2026-09-02
Board		Page	P1
Drawn		Inverted_MAML_6x6	
Reviewed			
		Version / Size / Page 1 Total 1	
EasyEDA		V1.0 / A4	EasyEDA.com



Schematic	fpga_t120_etc2	Create at	2026-08-10
		Update at	2026-08-10
Board		Page	P1
Drawn		Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4
		Page 1 Total 1	
		EasyEDA.com	


1.2V
3.3V
GND
U15CBB23.1
U15CBB23.2
T20F256C3
T20_CONDONE
T20_CRESET_N
FL_CS
FL_CSO
FL_DI
FL_DO
FL_TDI
FL_CLK
FL_IO3
FL_IO2
T20_TDO
T20_TMS
T20_TCK
T20_TDI
NSTATUS
TEST_N
SS_N
CBUS0
CBUS1
CBUS2
T20_IO_1
T20_IO_2
T20_IO_3
T20_IO_4
T20_IO_5
T20_IO_6
T20_IO_7
T20_IO_8
T20_IO_9
T20_IO_10
T20_IO_11
T20_IO_12
T20_IO_13
T20_CDI3
T20_CDI2
T20_CLK9
T20_CLK8
HEAT_1_EN
HEAT_2_EN
HEAT_3_EN
HEAT_4_EN
5.0V
U93CBB23
U118CBB23
U19CBB23
X1CBB23
W25Q64JVZPIQ
APX8836-31SA-7
PZ2.54-2*4
50MHz
BOOT_MODE
Width
x1
x2
x4
(Table 5, SPI Hardware Settings)
Schematic
T20F256_FPGA
Create at
2026-08-31
Update at
2026-08-31
Board
Page
P1
Drawn
Reviewed
Inverted_MAML_6x6
Version
V1.0
Size
A4
Page 1 Total 1
EasyEDA
EasyEDA.com



Hadamard coding signal strength increade. updated 2026-07-18 to new values

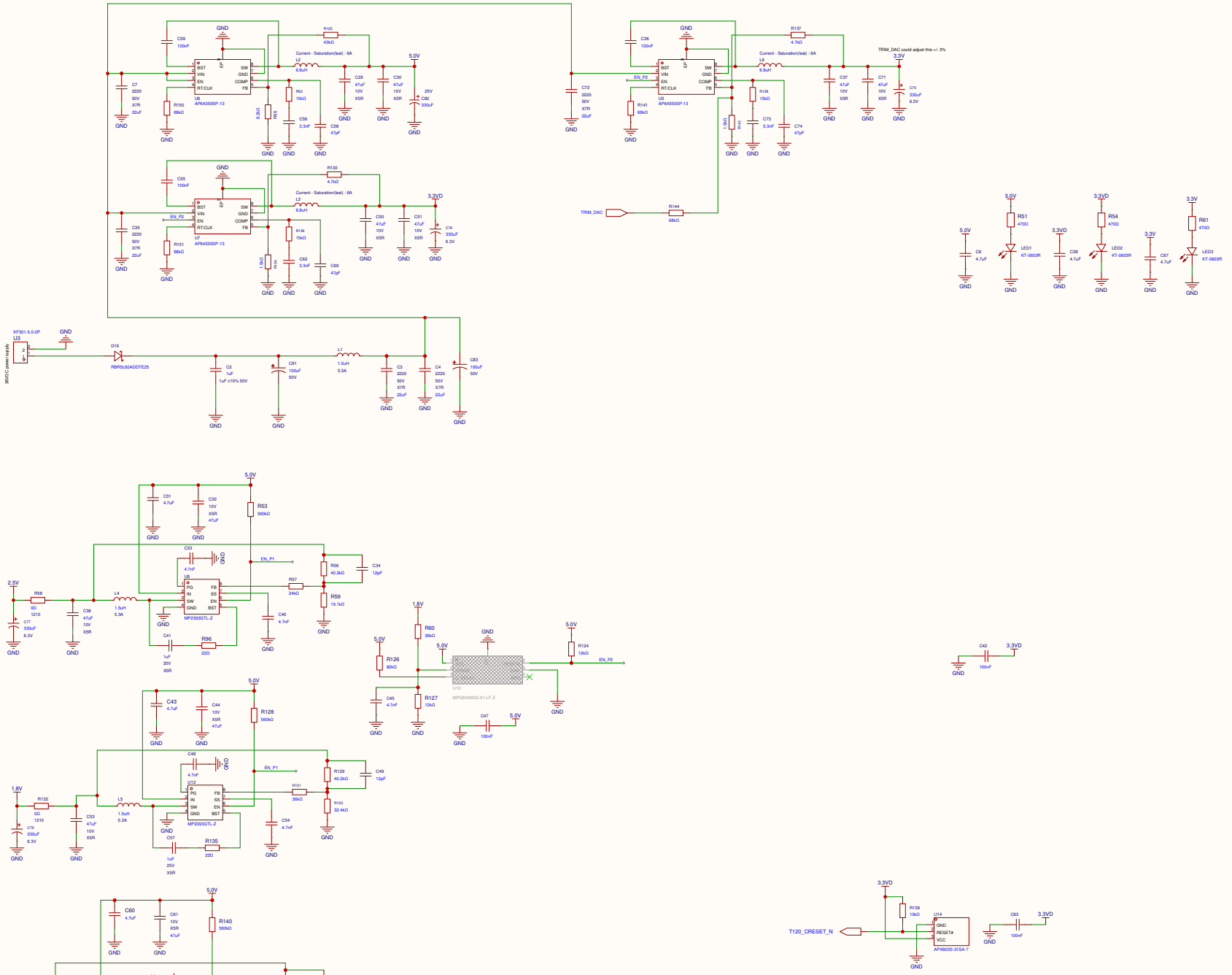
use the 2.2nF capacitors and still settle under 2 µs.

- $R_1 = \mathbf{110\ \Omega}$
- $R_2 = \mathbf{110\ \Omega}$
- $R_{series} = \mathbf{60\ \Omega}$ (or 56 Ω standard)

Your new Thevenin equivalent resistance drops to roughly **28 Ω**.

- **New Time Constant (τ):** $28\ \Omega \times 13.2\ \text{nF} = \mathbf{370\ ns}$
- **New Settling Time (5τ):** $5 \times 370\ \text{ns} = \mathbf{1.85\ \mu s}$

Schematic	Inverted_MAML	Create at	2026-07-20
		Update at	2026-07-20
Board	Analog_Matrix	Page	Haramard_resistor_nets
Drawn		Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 4 Total 5
EasyEDA		V1.0	A4	EasyEDA.com



Schematic	Inverted_MAML	Create at	2026-08-19
		Update at	2026-09-03
Board	Analog_Matrix	Page	Power Supply
Drawn		Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 5 Total 5
EasyEDA		V1.0	A4	EasyEDA.com